


# AURUM IQ

## Commodity Derivatives Intelligence Platform

### TEAM

**Team name: TimelessXO**

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# PROBLEM STATEMENT & THEME

**Commodity Derivatives Intelligence: four MCX gold contracts track the same metal but differ in size, quote basis, purity and expiry.**

## 1 Problem

- GOLDM (100 g, 995), GOLDTEN (10 g), GOLDGUINEA (8 g), GOLDPETAL (1 g) differ in unit, quote basis and purity
- Raw price gaps mix real mispricing with contract specs, expiry and bad data
- Traders and analysts lack a transparent, spec-aware way to separate the two

## 2 Impact

- Analysts misread normal spec differences as "opportunities"
- Backtests that ignore costs and liquidity overstate edges
- Decisions rest on unexplained charts rather than traceable evidence

## 3 Theme alignment

- Commodity derivatives intelligence for traders, researchers, brokers and analysts
- Data-driven, explainable analytics, software-only delivery
- Decision support, not guaranteed-profit prediction

# SOLUTION

**AURUM IQ normalises MCX gold futures onto a verified common basis, flags unusual relative prices, and tests whether they would have survived realistic costs.**

### Spec-aware normalisation

Size, quote basis and purity handled per contract; original and normalised price kept

### Data-quality gate

Validates returned date, strips padded symbols, flags gaps and stale prices; tracks by expiry, not near-month series

### Relative-value lab

Spread, rolling z-score, percentiles and robust (median/MAD) anomaly flags

### Cost-aware backtesting

Walk-forward replay, no lookahead, net of costs, thin-day liquidity, separated from gold price moves

### Evidence-grounded AI

Explains alerts only from computed results; works with no LLM key

### Transparent provenance

Every screen labels data as real, sample or synthetic

**Unique value: it can reject false signals and show why an attractive spread is not tradable.**

# ARCHITECTURE

**Ingest**
CSV / bhavcopy upload, validation, provenance

**Metadata**
Contract specs with effective dates

**Normalise**
Common basis, formula + source logged

**Analytics**
Spreads, z-score, percentile, anomalies

**Backtest**
Walk-forward, costs, baselines

**API + UI**
FastAPI to Next.js dashboard

## Tech stack

- Next.js, TypeScript, Tailwind, Recharts
- FastAPI, pandas, NumPy, SciPy
- PostgreSQL / SQLite, pytest
- Docker for reproducibility

## Data flow

- Raw file to validated observations
- to normalised prices (original kept)
- to spreads and alerts
- to backtest runs and exports

## APIs / data sources

- MCX Daily Bhavcopy: Symbol, Date, Expiry, OHLC, Volume, Open Interest; no API key
- Optional LLM API for explanations; app works without it
- No live exchange link needed for demo

# TARGET USERS, USE CASE & IMPACT

## Target users

- Commodity traders and derivatives researchers
- Broker and fintech research desks
- Students and quant analysts studying futures relative value

## Use case

- Load MCX Bhavcopy, pick two gold contracts
- See normalised spread, alerts and data warnings
- Test the signal net of costs before trusting it

## Expected impact

- Fewer false "mispricing" calls from spec differences
- Evidence-backed decisions with reproducible results
- Honest "no edge" answers instead of overfit charts

## Existing alternatives

- Charting terminals, broker dashboards, generic backtesters
- Mostly show raw prices, not spec-aware normalisation
- Landscape to be verified before claiming uniqueness

# IMPLEMENTATION ROADMAP & FUTURE SCOPE

## 1 Phase 1: 18-24 Oct

Bhavcopy ingest, data-quality checks, contract metadata, normalisation

## 2 Phase 2: by 27 Oct

Spread and z-score alerts, walk-forward backtest net of costs, demo video + GitHub

## 3 Phase 3: 20-21 Nov

Dashboard polish, evidence-grounded explanations, expiry-aware analysis

## 4 Future: coverage

Silver, crude and base metals, plus more exchanges

## 5 Future: data

Licensed intraday and bid-ask data for execution-realistic tests

## 6 Future: selective ML

Only where a leakage-safe out-of-sample task justifies it

# EXTRA: INNOVATION & UNIQUENESS

## What is new

- Spec-aware normalisation across four gold contracts
- Data validation runs before any signal is shown
- Every alert links to its calculation and limits

## Guardrails built in

- Chronological split, no lookahead
- Costs, liquidity and expiry rules always applied
- No-trade baseline must be beaten after costs

## Results are reported as

- Profitable, unprofitable or inconclusive
- Too few observations or poor data quality
- Signals that vanish after costs

## Disclaimer

- Research and decision support only
- Historical results do not guarantee future returns
- An anomaly is not automatically a tradable opportunity

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# Thank you!